Problem Chosen	2026	Team Control Number
C	MCM/ICM Summary Sheet	2607504

Deciphering the Invisible Vote: An Exploration of Evaluation Bias and Rule Fairness

Summary

Dancing with the Stars uses a dual evaluation scheme: judges' technical scores and fan votes. While this design increases suspense, it can also invite fairness concerns when the two components are unevenly weighted. Because vote counts are confidential and therefore latent, we model the dynamics of fan support to illuminate this "black box" and to identify a principled trade-off between technical excellence and public preference.

To address the unobservability of fan voting, we formulate a **partially observable dynamical system** for contestants' latent vote shares and solve it via particle filtering. The resulting model achieves a **Top-3 accuracy of 79.19%** and a **Top-1 accuracy of 37.92%** in forward prediction; in posterior backtesting, it reproduces **82.20%** of historical eliminations, indicating strong fidelity to observed outcomes. To quantify estimation confidence, we construct confidence intervals based on the posterior standard deviation of vote shares: eliminated contestants exhibit substantially tighter intervals than survivors, and overall uncertainty increases as the competition progresses.

Subsequently, we employed a **Counterfactual Analysis** framework to simulate potential outcomes under different rules, revealing significant heterogeneity between different scoring rules. The probability of the Rank Rule and Percentage Rule yielding identical results is merely **38.2%**. To dissect the directional inclination of each rule, we constructed statistical metrics for bias quantification. The results demonstrate that the Percentage Rule exhibits a pronounced "**Pro-Fan Bias**," whereas the Rank Rule effectively constrains the marginal influence of fan votes through a "**Capping Effect**".

To analyze controversial cases, we constructed the Robbed Score and Fan-Carried Score to quantify two types of controversy: High-Skill Dropouts and Low-Skill High-Rankers. Utilizing these metrics, we precisely identified the four controversial contestants mentioned in the problem statement and analyzed their performance trajectories. Our findings reveal that for technically skilled contestants with limited fan bases, the Percentage Rule causes an average decline of 0.32 in their expected ranking. In contrast, the Judge Save mechanism serves as a critical corrective tool for these candidates, who are otherwise prone to falling into the elimination zone. Ultimately, after evaluating the directional biases of various rules regarding fan and judge influence, we recommend the **Rank with Judge Save rule** as the optimal framework for ensuring competition integrity.

To systematically quantify the contributions of various influential factors, we established a **Linear Mixed-Effects Model** (LMM) framework. Our results reveal that the impact of Professional Partners is statistically negligible; instead, competition outcomes are predominantly governed by the intrinsic attributes of the celebrities. Actors emerged as the most competitive, consistently securing superior fan votes and judge scores. Furthermore, the data uncovers a significant **divergence in preferences:** fans exhibit a strong propensity to vote for Politicians, whereas judges demonstrate greater recognition for the performances of Reality Stars and Musicians.

Finally, leveraging our quantitative metrics, we constructed the Adaptive Concave Diminishing-returns Weighted Bottom-3 System (ACDW-B3). This mechanism **dynamically amplifies the weighting of judges' scores** during periods of significant divergence between professional assessment and audience preference. Consequently, it allows the program to manufacture highly engaging "conflicts of opinion" and "dramatic tension," while safeguarding the professional integrity of the competition results.

Keywords: Particle Filtering, Latent Variable Inference, Counterfactual Inference, Linear Mixed-Effects Model

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1 Introduction

1.1 Restatement of the Problem

In the global entertainment market, competitive reality shows have built a large audience base by combining professionalism with entertainment value. As a prime example of this genre, *Dancing with the Stars*—adapted from the British show *Strictly Come Dancing*—pairs celebrities with professional dancers and has achieved extensive global influence. The show uses a dual scoring system that integrates expert judge scores and fan votes to determine weekly eliminations. This design requires a necessary trade-off between competitive fairness and audience engagement.

Based on the comprehensive dataset from Seasons 1–34 of Dancing with the Stars—including contestant profiles, judges' scores, and elimination results—this study focuses on predicting latent fan vote shares, optimizing scoring mechanisms, and quantifying influential factors. Specifically, we addressed the following four tasks:

- **Task 1: Estimation of Latent Fan Vote Shares.** We will formulate a mathematical framework to predict the unobservable fan vote shares for each contestant on a weekly basis. To ensure the model's reliability, we aim to develop specific consistency metrics to validate our estimates against historical elimination outcomes and design quantifiable indicators to measure the certainty of these latent variable predictions.
- **Task 2: Evaluation of Scoring Rules & Controversy Analysis.** By integrating the estimated fan votes, we plan to conduct a comparative analysis of the Rank Rule and Percentage Rule to dissect their directional biases toward fan vote share versus judges' score. We intend to scrutinize controversial contestants—where significant divergence exists between judges and fans—to determine if alternative rules would alter their elimination fates. Additionally, we will investigate the corrective impact of the "Judge Save" mechanism to recommend an optimal voting framework.
- **Task 3: Quantification of Influential Factors.** Using the dataset augmented with estimated fan votes, we propose to establish statistical models to disentangle the impacts of Professional Partners and Celebrity Characteristics on competition outcomes. We seek to quantify the influence of these factors and contrast their differing effects on judges' technical scores versus fan voting share.
- **Task 4: Mechanism Design for Fairness and Entertainment.** Building on the preceding analyses, we aim to design a novel hybrid scoring system that effectively balances professional fairness with entertainment value. We will provide a detailed mathematical justification to persuade producers that this system can maintain dramatic tension without compromising the competitive integrity of the show.

1.2 Literature Reviews

The optimization of scoring and voting aggregation mechanisms has been the subject of extensive academic research. Molinero et al. (2015) utilized hierarchical clustering and spatial network analysis to demonstrate that a single aggregation method may inadvertently suppress the voices of audiences in specific regions [1]. Focusing on algorithmic fairness, Dwork et al. (2012) proposed the principle of "fairness through awareness" and employed similarity metrics to constrain disparate treatment [2]. Aird et al. (2024) explored dynamic fairness-aware recommendation through a multi-agent social-choice framework [3]. Furthermore, Collins et al. (2019) established econometric models to identify potential sequential performance bias in expert evaluation [4].

1.3 Our Work

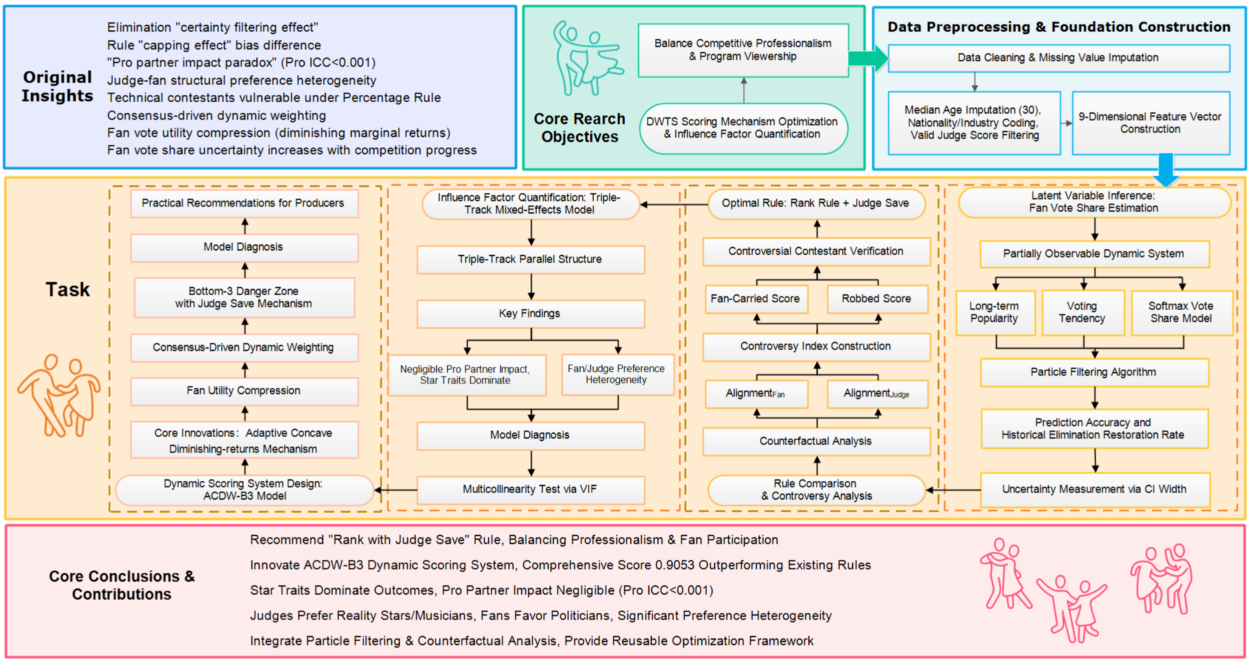

Figure 1: Our Work Overview

2 Preparation for Modelling

2.1 Assumptions

Assumption 1: Markovian Property of Public Memory. We assume that the evolution of fan support follows a Markov process, where a contestant's current popularity depends primarily on the momentum retained from the previous week and their performance in the current week, rather than directly relying on their entire historical trajectory.

Justification: Due to the fast-paced nature of reality TV and social media, the audience's attention span is short. Most viewers do not retrospectively evaluate a contestant's performance from distant weeks; instead, their voting decisions are driven by immediate impressions and the lingering memory from the most recent episode.

Assumption 2: Existence of Judges' Influence We assume that judges' scores exert a steering effect on fan voting behavior, manifesting through phenomena such as "sympathy votes" or the "bandwagon effect." This influence is modeled as a linear superposition within the vote generation mechanism.

Justification: Professional critiques and scores are revealed prior to the public voting window, inevitably priming audience preferences. Furthermore, modeling this influence as a linear addition captures the core mechanism while maintaining model interpretability and avoiding unnecessary non-linear complexity.

Assumption 3: Elimination Results as the Sole Ground Truth. We assume that an eliminated contestant necessarily possessed the lowest (or one of the lowest) combined score for that week.

Justification: From the perspective of operational logic, commercial reality competitions must maintain fundamental rule fairness to sustain audience trust. To ensure the show's credibility and public appeal, the elimination mechanism must intrinsically rely on the quantitative ranking of the contestants' combined performance scores.

2.2 Data Preprocessing

We conducted comprehensive data preprocessing on the raw dataset, primarily focusing on **missing value handling**, **feature encoding**, and **score standardization**.

2.2.1 Data Cleaning and Missing Value Imputation

The raw dataset exhibits discrepancies in field completeness and formatting. To address these inconsistencies, we implemented the following data processing strategies:

Handling Missing Data: We filled missing "Celebrity Age" values with the median (30) and set missing "Home Country" to "Unknown." For the "placement" field, we assigned a value of 99 to indicate an unranked status.

Determination of Elimination Status: We identified eliminated contestants based on explicit weekly result records. Voluntary withdrawals were excluded to ensure the analysis focused strictly on competitive performance outcomes.

Refining Judges' Scores: To address "N/A" or 0 values, we filtered the data to retain only valid scores greater than 0. A contestant's weekly score was then determined by the arithmetic mean of all valid judges' scores for that week.

2.2.2 Feature Engineering

To prepare the data for our model, we processed the raw features and created a final **9-dimensional feature vector**.

Age Standardization: To fix scale differences between variables, we standardized age using the following formula:

$$A_{norm} = \frac{Age - 35}{15} \tag{1}$$

Nationality Indicator: We created a simple binary variable, I_{US}, to show if a contestant is from the United States. We set this to 1.0 for US contestants and 0.0 for others.

Industry One-Hot Encoding: We grouped contestant careers into seven categories: Actor, Athlete, Singer, TV Personality, Model, Comedian, and Other. We used **One-Hot Encoding** for these categories. This helps the model see how a contestant's job affects fan votes.

2.2.3 Post-Processing Data Summary

After data cleaning, we obtained a final valid sample covering **34 seasons** and **421 contestants**. The dataset records **298 elimination events**, averaging **12.4 contestants per season**.

2.3 Notation

We define the primary mathematical notations used throughout this paper as follows:

Table 1: Notation

Symbol	Description
$J_{i,t}$	Judge score of contestant i at week t
$\pi_{i,t}$	Fan vote share of contestant i at week t
$\hat{\pi}_{i,t}$	Posterior estimate of the fan vote share for contestant i
$S_{i,t}$	Survival score for contestant i at week t
R	Scoring rule
$E_{real,t}$	Real elimination set at week t
$Alignment_{Fan}(R)$	Fan alignment score under rule R
$Alignment_{Judge}(R)$	Judge alignment score under rule R
S_1, S_2	Controversy metrics: Fan-Carried / Robbed
$Y^J_{i,t}$, $Y^F_{i,t}$	Dependent variables in mixed-effects models (judge/fan)
X_{Traits}	Contestant trait vector
β^J, β^{F1}, β^{F2}	Fixed-effect coefficients
u_{Pro}, u_{Star}	Random intercepts for professional partners / celebrities
ICC_{Pro}	Variance contribution of professional-partner random effects
$f_{i,t}$	Compressed vote utility after a concave transform
p	Compression factor
ρ_t	Spearman correlation between judges' and fans' rankings
λ_t	Dynamic weight on judges scores
w^k_t	Weight of particle k at week t in the particle filter

3 Task 1: Estimation of Latent Fan Vote Shares

3.1 Problem Analysis

First, we define the nature of the problem: the weekly judges' scores $J_{i,t}$ and the final elimination outcomes $E_{real,t}$ are observable, whereas the actual fan voting data remains unobservable. Our objective is to reconstruct the latent fan vote share $\pi_{i,t}$ for each contestant based on these known inputs; consequently, this constitutes a **Latent Variable Inference** problem.

Furthermore, given that a season comprises a sequence of weekly competitions, a contestant's fan vote share depends not only on their current performance but also on the momentum accumulated in previous weeks, exhibiting strong **temporal dependency**. Therefore, the voting process should be modeled as a dynamic system. Synthesizing the characteristics of "partial observability" and "dynamic evolution," we classify this problem as a **Partially Observable Dynamic System**.

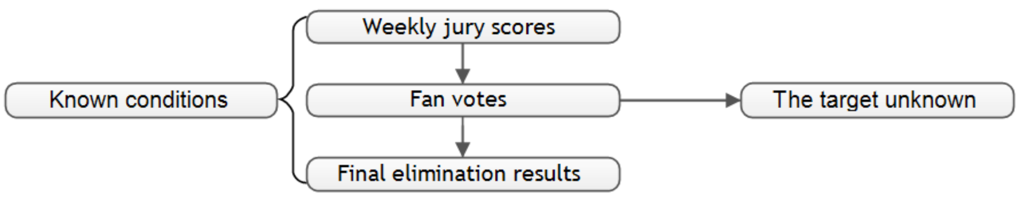

Figure 2: Flowchart of Solving the Model

3.2 Model Establishment and Solution

3.2.1 Modeling Long-Term Popularity and Latent Vote Shares

Having defined the problem structure, we proceed to formulate the mathematical framework for estimating the latent fan vote shares. We posit that a contestant's vote share is governed by two primary components: their **long-term popularity** (cumulative momentum) and the **short-term voting tendency** of the audience.

The evolution of a contestant's long-term popularity, denoted as $\mu_{i,t}$, is modeled as follows:

$$\mu_{i,t} = \mu_{i,t-1} + \kappa \cdot \mathbb{I}(|\tilde{J}| > \delta) \cdot \tilde{J} + \eta_t \tag{2}$$

The contestant's long-term popularity state at time t, denoted as $\lambda \equiv \mu_{i,t}$, evolves from the state at the previous time step $\mu_{i,t-1}$ with the addition of a process noise $\eta_t \sim (0, \sigma_\mu)$. We introduce a mechanism to capture "significant events"—instances where the absolute value of the Z-score standardized judge rating, $\tilde{J}$, exceeds a designated threshold δ. These events, representing either exceptional excellence or notable failure, exert a directional shock on the contestant's long-term popularity status. Specifically, $\tilde{J} > 0$ signifies an accumulation of popularity, whereas $\tilde{J} < 0$ indicates a decline, with the intensity of this shock being directly proportional to the magnitude $|\tilde{J}|$.

We define $x_{i,t}$ as the audience's short-term voting tendency for contestant i at week t. This tendency evolves dynamically, driven by the combined influence of a memory persistence term from the previous week, the current long-term popularity state $\mu_{i,t}$, and the immediate stimulus provided by the judges' score $J_{i,t}$. The state transition model is formulated as follows:

$$x_{i,t} = \rho \cdot x_{i,t-1} + (1 - \rho) \cdot (\mu_{i,t} + \gamma \cdot J_{i,t}) + \varepsilon_t \tag{3}$$

Here, $\rho \in (0, 1)$ represents the memory persistence coefficient for the voting tendency: a larger ρ implies that $x_{i,t}$ relies more heavily on the inertia from the previous week, whereas $(1 - \rho)$ governs the injection intensity of the current week's new signal (defined as $\mu_{i,t} + \gamma \cdot J_{i,t}$) into the state $x_{i,t}$. The parameter γ characterizes the influence of the judges' score $J_{i,t}$ on the audience's voting tendency.

This configuration incorporates both the long-term popularity state $\mu_{i,t}$ and the short-term perturbations driven by the current week's scoring. By doing so, we characterize the dual dynamics of voting tendencies—capturing both secular trends and transient fluctuations—within a unified framework.

Consequently, the vote share for each contestant is determined using the Softmax function:

$$\pi_{i,t} = softmax(x_t) = \frac{exp(x_{i,t})}{\sum_k exp(x_{k,t})} \tag{4}$$

3.2.2 Rule Analysis and Calculation of Survival Scores

Having derived the contestants' vote shares, we proceed to compute their scores based on the rules. The competition framework is categorized into three core paradigms; the definitions and computational logic for each phase are summarized in the table below:

Table 2: Analysis of Scoring Rules

Season Range	Rule Paradigm	Computational Logic
S1-S2	Rank Rule	Add judge rank + vote rank
S3-S27	Percentage Rule	Weighted sum of judge scores% + vote%
S28+	Rank + Judge Save	Rank sum + Judge Save

To adjudicate the advancement or elimination of contestants, we define the **Survival Score** as the pivotal metric. Different rule paradigms necessitate distinct computational formulas for this score. Under the Percentage Rule, the final score is composed of equal weights (50%) assigned to the proportion of judges' scores and the fan vote share. Let S_i denote the final Survival Score for contestant i. Here, J_i represents the weekly score awarded by the judges, and π_i represents the contestant's concurrent vote share. The core computational formula is defined as follows:

$$S_i = (\frac{J_i}{\sum_k J_k}) + \pi_i \tag{5}$$

In contrast, under the Rank Rule, let $Rank_{J_i}$ and $Rank_{\pi_i}$ denote the contestant's ranking derived from the judges' scores and the vote share, respectively. Since a numerically lower rank indicates superior performance, the sum of ranks functions as an inverse metric; consequently, we adopt the negative sum of ranks as the core computational variable.

Furthermore, actual competitions present the complexity of tie-breaking scenarios. According to the producer's official explanation, in the event of a tie in the aggregate rank score, the contestant with the lower fan vote count is eliminated. To ensure high consistency between our model and this official elimination logic, we incorporate a tie-breaking perturbation term, $\xi \cdot \pi_i$, into the survival score formula. Here, ξ serves as a perturbation coefficient designed to resolve adjudication ambiguities when rank sums are identical:

$$S_i = -(Rank_{J_i} + Rank_{\pi_i}) + \xi \cdot \pi_i \tag{6}$$

Furthermore, *Dancing with the Stars* occasionally features weeks where no elimination occurs; instead, the scores from the current week are aggregated with those of the subsequent week to adjudicate elimination. To address such cases, our model adopts differentiated accumulation strategies contingent upon the specific rule paradigm:

Accumulation of Judges Scores: Across all seasons, in the event that no elimination occurs in week t, the judges' score for the subsequent week, $J_{i,t+1}$, automatically aggregates the scores from the preceding unresolved weeks:

$$J_{i,accumulated} = \sum_{T \in \text{Non-elimination Window}} J_{i,T} \tag{7}$$

Accumulation of Vote Shares: Percentage Rule (S3-S27): We employ a strategy of direct share summation. Given that the fundamental mechanism of this rule is the addition of the percentage of total points and the percentage of total votes, it is logical to directly accumulate the contestants' vote shares.

Rank Rule (S1-2, S28+): We utilize rank-point accumulation. Under this rule, the model aggregates the contestant's weekly ranking points rather than the raw vote shares.

3.2.3 Model Establishment

To solve the proposed model, we employ Particle Filtering (Sequential Monte Carlo). For latent variable inference problems of this nature, the four prevailing methodologies are Regression, the Kalman Filter, Markov Chain Monte Carlo (MCMC), and Particle Filtering. A comparative analysis of these methods is presented below:

Table 3: Comparison of Model Solution Methods

Method	**Key idea**	**Main caveat (here)**	**Fit**
Regression	Static mapping.	Lacks temporal dynamics (memory).	Low
Kalman filter	Recursive; linear-Gaussian.	Poor performance with discrete elimination events.	Low
MCMC	Posterior sampling.	Offline processing; too slow for sequential updates.	Medium
Particle filter	Sequential; nonlinear.	Computationally intensive.	High

The computational workflow employed to solve the model is illustrated in the figure below:

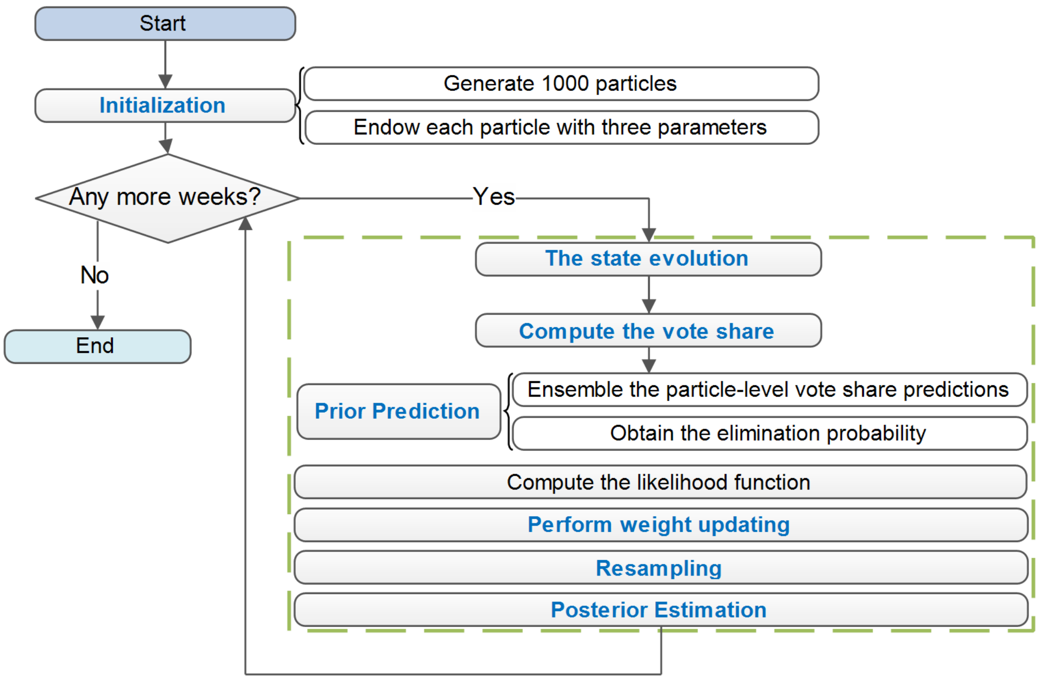

Figure 3: Uncertainty Heterogeneity Analysis

We commence the algorithm by initializing an ensemble of $N = 1000$ particles. Each particle k is characterized by a state vector comprising four parameters: μ_0^k, x_0^k, w_0^k, and S_t^k, which correspond to the initial long-term popularity, initial short-term popularity, initial weight, and initial survival score, respectively.

For each particle at every weekly time step, we propagate the state variables through the evolution equations to yield the predicted estimates μ_t^k and x_t^k. Subsequently, the short-term vote share is derived via the Softmax function: $\pi_t^k = softmax(x_t^k)$.

We then perform a prior prediction by aggregating the vote shares across the particle ensemble to estimate the probability of elimination:

$$P(i \text{ is eliminated}) = \sum_k w_k \cdot \mathbb{I}[S = \min(S^k)] \tag{8}$$

Next, we compute the likelihood L_t^k based on the observed elimination results. The particle weights are updated recursively according to $\tilde{w}_t^k = w_{t-1}^k L_t^k$ and subsequently normalized:

$$w^k = \frac{\tilde{w}^k}{\sum_i \tilde{w}^i} \tag{9}$$

A resampling step is then executed to mitigate particle degeneracy, where particles are drawn with replacement with probabilities proportional to their normalized weights w^k.

Finally, the posterior estimate of the contestant's vote share is computed as the weighted average of the particles:

$$\hat{\pi}_i = \sum_k w^k \cdot \pi_i^k \tag{10}$$

The methodology for computing the likelihood is detailed below. The objective of the model is to derive the posterior vote shares based on observed data. Under the assumption of model validity, the *Survival Score* of advancing contestants must theoretically exceed that of eliminated contestants. Predicated on this logic, we construct the corresponding likelihood statistics.

a. Baseline Elimination Likelihood (Pre-Season 28)

For cases involving a single elimination event, the likelihood function is defined as:

$$L = \prod_{j \in Survivors} \sigma(\alpha \cdot (S_j^k - S_e^k)) \tag{11}$$

In cases where multiple contestants are eliminated simultaneously, the likelihood generalizes to the product of pairwise comparisons between all survivors and all eliminated candidates:

$$L = \prod_{e \in Eliminated} \prod_{s \in Survivors} \sigma(\alpha \cdot (S_s^k - S_e^k)) \tag{12}$$

Here, α serves as the *discriminative power parameter*. It is designed to increase monotonically over time, signifying that the boundary between survival and elimination becomes increasingly distinct and rigorous as the competition progresses into later stages.

b. Judges' Save Likelihood - Post-Season 28

From Season 28 onwards, the elimination mechanism evolves into a two-stage probabilistic event. The first stage involves falling into the "Bottom 2," a status contingent upon the Survival Score, $S_{i,t}$. The second stage is the "Judges' Decision," which is determined by technical merit. Consequently, the likelihood function is formulated as follows:

$$L = \underbrace{\left[\prod_{j \in S_{safe}} \sigma\left(\alpha \cdot (S_j - \max(S_e, S_s))\right) \right]}_{\text{Stage 1: into Bottom 2 Probability}} \times \underbrace{\sigma\left(\beta \cdot (J_s - J_e)\right)}_{\text{Stage 2: Judges' Save Probability}} \tag{13}$$

Here, the second stage is modeled using a Sigmoid function dependent solely on the differential in technical scores.

3.3 Model Accuracy and Validation

To assess the predictive fidelity and internal consistency of our model, we employ three distinct metrics: **Top-N Hit Rate**, **Posterior Consistency Probability**, and **MAP (Maximum A Posteriori) Match Rate**. Specifically, the Posterior Consistency Probability and MAP Match Rate serve as diagnostics for model fitting during the training phase (in-sample validation), while the Top-N Hit Rate evaluates the model's performance in forward prediction (out-of-sample validation).

The Posterior Consistency Probability quantifies the aggregate weight of particles that successfully predict the correct elimination outcome following the weight update step. It is mathematically formulated as:

$$P_{consistency} = \sum_{k=1}^{K} w_t^k \cdot \mathbb{I}(Sim(R, J_t, \pi_t^k) \in E_{real,t}) \tag{14}$$

Here, $E_{real,t}$ denotes the set of contestants actually eliminated in week t. The function $Sim(R, J_t, \pi_t^k)$ represents the theoretical elimination result derived under the specific competition rules R, given the Judge Score J_t and the latent vote share $\pi^{(k)}$ associated with the k-th particle.

To evaluate the model's reconstructive capacity over the entire historical span (Seasons 1-34), we introduce the **MAP Match Rate**. This metric essentially represents the empirical probability that the model successfully reproduces the historical ground truth.

$$\text{Average MAP Match Rate} = \frac{1}{T} \sum_{t=1}^{T} \underbrace{\mathbb{I}\left(\text{Sim}(R, J_t, \hat{\pi}_t^{MAP}) \in E_{real,t}\right)}_{\text{Indicator of correct prediction at week } t} \tag{15}$$

Where $\hat{\pi}_t^{MAP}$ denotes the expected value of the posterior distribution, approximated by the weighted average of the particle ensemble:

$$\hat{\pi}_{i,t}^{MAP} = \mathbb{E}[\pi_{i,t} | D_{1:t}] \approx \sum_{k=1}^{K} w_t^k \cdot \pi_{i,t}^k \tag{16}$$

$$\hat{\pi}_{i,t}^{MAP} = \mathbb{E}[\pi_{i,t} | D_{1:t}] \approx \sum_{k=1}^{K} w_t^k \cdot \pi_{i,t}^k \tag{17}$$

To further evaluate the model's predictive efficacy regarding elimination events, we introduce the **Top-N Hit Rate** metric. Its calculation is defined as follows:

$$Top - N = \frac{1}{T} \sum_{t=1}^{T} \mathbb{I}(Rank_t(E_{real,t}) \leq N) \tag{18}$$

Here, $Rank_t(E_{real,t})$ denotes the rank assigned to the contestant actually eliminated in week t. This ranking is derived by sorting all contestants in descending order based on their model-predicted elimination probabilities. Consequently, this metric serves to assess whether the model successfully anticipated the correct outcome prior to the actual occurrence of the elimination.

Through an exhaustive grid search encompassing 4,320 distinct combinations within the full parameter space, we identified several parameter configurations of particular significance, as detailed below:

Table 4: Results of Parameter Grid Search

Category	ρ	γ	δ	α
Overall Best	0.9	0.0	1.2	7.0
Max Accuracy	0.8	0.7	1.8	7.0
Max Precision	0.7	0.8	1.8	7.0

The hyperparameters corresponding to the "Overall Best" configuration were adopted for the final model. The resulting performance metrics are presented in the following table:

Table 5: Model Performance Evaluation

Metric	**Hit Rate**	**Random Baseline**
Top-1	37.92%	$\sim$ 10%
Top-3	79.19%	$\sim$ 30%

It is evident that the model's metrics are significantly superior to the random baseline. This substantial margin of improvement demonstrates the validity and predictive efficacy of our proposed model.

Furthermore, the accuracy metrics across different seasonal eras are presented below:

Table 6: Accuracy Across Rule Regimes

Rule Paradigm	**Average Hit Rate**
Percentage (S3-S27)	80.40%
Rank (S1-2, S28+)	77.25%

The comparable performance observed under both rule paradigms indicates that the model has correctly internalized the distinct computational logic governing each era, demonstrating its structural robustness.

3.4 Analysis of Model Certainty and Heterogeneity in Uncertainty

Relying solely on point estimates is often insufficient for a comprehensive assessment of model reliability. Leveraging the non-parametric nature of Sequential Monte Carlo (SMC) methods, which are based on particle sampling, we can not only derive the expected value of fan vote shares but also fully reconstruct their posterior Probability Density Functions (PDF). In this section, we establish metrics for quantifying certainty based on particle distributions and analyze the heterogeneity of uncertainty across different competitive states.

3.4.1 Construction of Certainty Metrics

To quantify the credibility of the estimation results, we construct statistical metrics based on the particle ensemble following the resampling step at week t.

First, we calculate the **Weighted Posterior Mean** for each contestant at week t to serve as the point estimate for the vote share. The calculation is formulated as follows:

$$\hat{\pi}_{i,t} = \mathbb{E}[\pi_{i,t}|\mathcal{D}_{1:t}] \approx \sum_{k=1}^{K} w_t^{(k)} \cdot \pi_{i,t}^{(k)} \tag{19}$$

Second, we compute the **Weighted Posterior Standard Deviation** to measure the degree of dispersion within the distribution:

$$\sigma_{i,t} = \sqrt{\sum_{k=1}^{K} w_t^{(k)} \cdot (\pi_{i,t}^{(k)} - \hat{\pi}_{i,t})^2} \tag{20}$$

Predicated on the normal approximation assumption under the Central Limit Theorem, we construct the 95% **Confidence Interval (CI)**:

$$CI_{i,t} = [max(0, \hat{\pi}_{i,t} - 1.96\sigma_{i,t}), min(1, \hat{\pi}_{i,t} + 1.96\sigma_{i,t})] \tag{21}$$

Finally, we define the **Confidence Interval Width (CI Width)** as the core metric for model uncertainty:

$$Uncertainty_{i,t} = CI_{i,t}^{upper} - CI_{i,t}^{lower} \tag{22}$$

A wider interval indicates substantial uncertainty in the estimation.

3.4.2 Heterogeneity of Uncertainty: Survivors vs. Eliminated Contestants

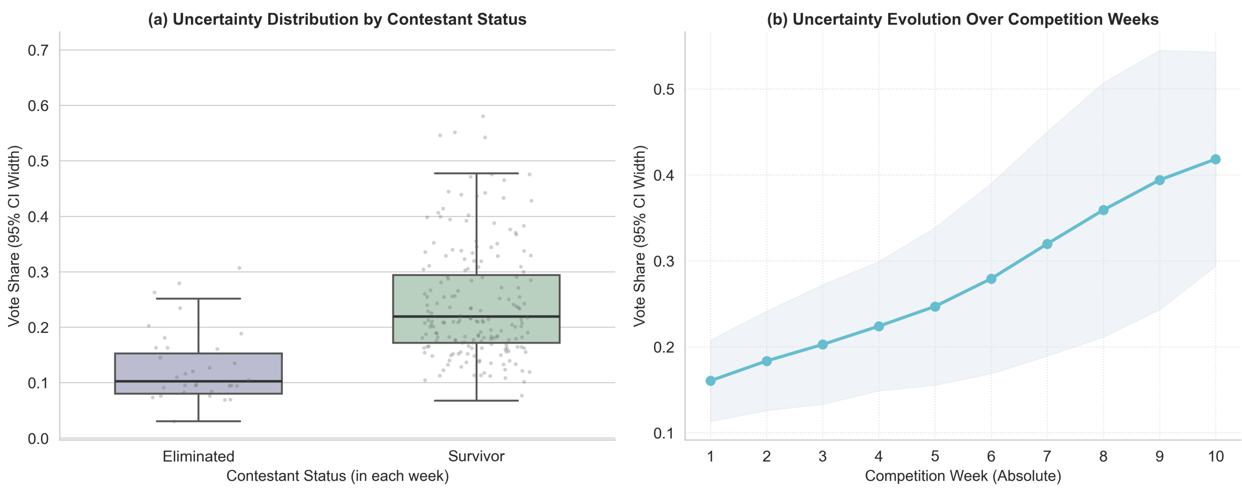

Figure 4: Analysis of Uncertainty Heterogeneity

A salient phenomenon emerges from the analysis: the model exhibits significantly higher estimation certainty for eliminated contestants compared to survivors. Specifically, the average **CI Width** for eliminated contestants is markedly narrower than that of their surviving counterparts.

This disparity acts as a response to the fact that an "elimination" constitutes an extremely stringent observational event. Theoretically, it mandates that the contestant's composite **Survival Score** must strictly fall below that of all other surviving competitors. Within the likelihood computation, this condition functions as a potent "filter": particles hypothesizing "high" or "moderate" vote shares for the eliminated candidate are assigned negligible weights or are systematically discarded.

Consequently, the posterior distribution of the remaining particles is forced to collapse into a constrained interval characterizing a "low vote share." Thus, the model achieves high determinism regarding the status estimation of eliminated contestants.

Concurrently, we observe a temporal trend where the uncertainty associated with *Vote Share* increases as the competition progresses. This is mechanically driven by the reduction in the number of contestants; as the field narrows, the expected Vote Share for each remaining contestant increases, naturally expanding the absolute variance and the width of the confidence intervals.

4 Task 2:Counterfactual Assessment of Scoring Rule

4.1 Problem Analysis

Given the estimated fan vote shares $\hat{\pi}_{i,t}$, the observed judge scores $J_{i,t}$, and the real elimination results, we formulate the following three primary objectives:

Objective 1: Compare and contrast the divergence in outcomes produced by distinct scoring rules.

Objective 2: Analyze the trajectories of controversial contestants under alternative rule.

Objective 3: Recommend an optimal scoring mechanism.

Objectives 1 and 2 necessitate the analysis of scenarios that did not empirically occur; consequently, we frame this as a problem of **Probabilistic Counterfactual Inference**. Furthermore, Objectives 1 and 3 involve evaluating the relative efficacy of different rules, requiring the construction of robust quantitative metrics for comparative assessment.

4.2 Comparative Analysis and Rule Selection

4.2.1 Computational Framework for Counterfactual Outcomes

To quantify the divergence in outcomes between the historical and alternative rule, we leverage the posterior particle ensemble derived in Task 1. This approach allows us to simulate the competition under counterfactual scenarios while preserving the estimated latent voter dynamics.

The procedure operates as follows: utilizing the posterior particles, we first extract the latent vote share π^k for each contestant. We then re-evaluate the Survival Score S^k_{new} and the corresponding ranking by applying the alternative rule R_{new}. Finally, we derive the elimination probabilities and outcomes specific to the new rule. The computation remains consistent with the methodology established in Task 1.

Under the alternative rule, the contestant with the lowest Survival Score is designated for elimination. Consequently, the probability of elimination for contestant i under rule R_{new} is formulated as the weighted aggregation of elimination events across the particle ensemble:

$$P(i \in E|R_{new}) = \sum w^k \cdot \mathbb{I}(i \in E^k) \tag{23}$$

$$Alignment_{Fan}(R) = \frac{\mathbb{E}_{k \leq K}[Rank_\pi(E^k)] - 1}{n_t - 1} \tag{24}$$

4.2.2 Analysis of Rule Divergence

Having derived the counterfactual outcomes under the alternative rule, we proceed to quantify the divergence between the two scoring regimes. We employ the **Result Agreement Rate** to measure the consistency of elimination outcomes across different rules. The seasonal agreement rate is formulated as follows:

$$P_t(E_{rank} = E_{pct}) = \sum_{k=1}^{K} w^k \cdot \mathbb{I}(E^k_{rank} = E^k_{pct}) \tag{25}$$

$$Agreement = \frac{1}{T}\sum_{t=1}^{T} P_t(E_{rank} = E_{pct}) \tag{26}$$

Here, P_t denotes the weekly result agreement probability, and $Agreement$ represents the aggregate seasonal consistency, calculated as the arithmetic mean of the weekly agreement rates.

Subsequently, we define the **Fan Alignment Score** to quantify the sensitivity of the results to audience voting. A higher value indicates that the fan vote exerts a more dominant influence on the outcome. The calculation is defined as:

$$Alignment_{Fan}(R) = \frac{\mathbb{E}_{k\le K}[Rank_\pi(E^k)] - 1}{n_t - 1} \tag{27}$$

Where the expectation operator $\mathbb{E}_{k\le K}$ represents the weighted average over the K particles, and the term $n_t - 1$ serves as a normalization factor for the ranking. The metric ranges from 0 to 1. A value of 0 implies the rule eliminated the contestant with the highest popularity (Rank 1), whereas a value of 1 implies the elimination of the contestant with the lowest popularity (last place).

Finally, we compute the **Judge Alignment Score**. A higher value signifies that technical merit (judges' scores) plays a more decisive role in the outcome. The formula is given by:

$$Alignment_{Judge}(R) = \frac{\mathbb{E}_{k\le K}[Rank_J(E^k)] - 1}{n_t - 1} \tag{28}$$

The computational comparison between the Rank Rule and the Percentage Rule is visualized below:

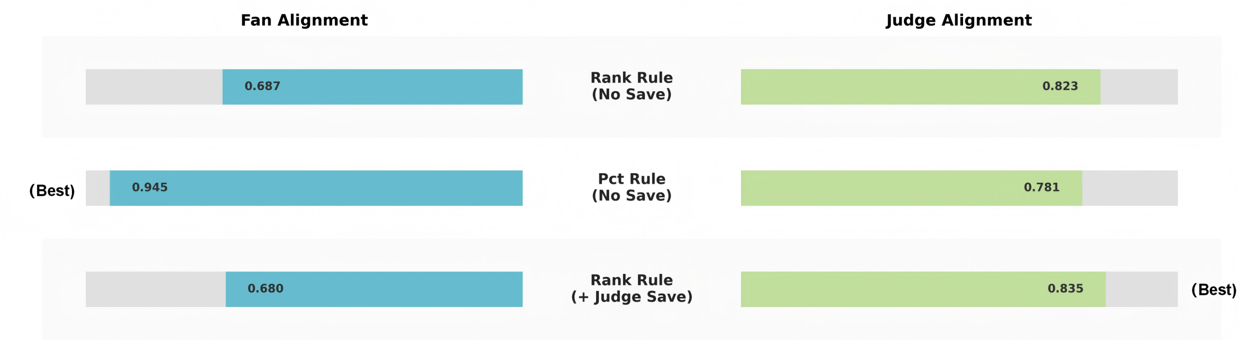

Figure 5: Comparative Analysis of Rule Consistency

Based on the quantitative data derived from our simulations, we draw the following critical conclusions:

Inherent Rule Disagreement: Our counterfactual simulations reveal a substantial divergence between the two rules. The seasonal result agreement rate between the Rank Rule and the Percentage Rule is merely 38.2%, indicating that in only 38.2% of cases do these distinct mechanisms yield identical elimination outcomes.

Bias Directionality: The Percentage Rule exhibits a pronounced **Pro-Fan Bias**. Its Fan Alignment Score ($Align_{Fan}$) reaches as high as 0.945, exceeding that of the Rank Rule (0.687) by a margin of 0.38. This disparity arises because the proportional nature of the rule allows the massive vote surpluses of "superstars" to directly inundate the deficits in Judge Score. Conversely, the Rank

Rule demonstrates superior **Pro-Judge**. Its Judge Alignment Score ($Align_{Judge}$) stands at 0.823, surpassing the Percentage Rule (0.781) by approximately 5%. By converting raw data into ordinal rankings, the Rank Rule effectively imposes a ”**Capping Effect**” on the marginal influence of fan votes, thereby increasing the probability that contestants with lower technical merit fall into the danger zone.

Furthermore, we computed the **Seasonal Result Agreement Rate** across distinct historical epochs, revealing significant heterogeneity in consistency over time:

S1-S2 (Rank Era): The global agreement rate reached a high of 73.2%. During this nascent phase, the differentiation among contestants was relatively low, which effectively masked the structural divergence between the scoring rules.

S3-S27 (Percentage Era): The agreement rate plummeted to 28.5%. This sharp decline indicates that the scoring mechanism during this era exhibited an extreme **Pro-Fan Bias**, resulting in a high frequency of “upsets” that defied the logic of a meritocratic ranking system.

S28+ (Modern Era): The agreement rate rebounded to 66.4%. Following the introduction of the **Judge Save** mechanism, the competition framework successfully re-established an equilibrium between technical merit and popular appeal.

4.3 Analysis of Controversies

4.3.1 Construction of Controversy Metrics

To quantify the degree of controversy associated with specific contestants, we construct a set of dedicated indicators. These metrics are designed to capture the **structural divergence** between “**professional evaluation**” and “**fan preference**.” This discordance manifests primarily in two distinct cases: contestants possessing high technical merit who suffer premature elimination, and contestants with low technical scores who nevertheless achieve superior rankings.

We define two metrics to quantify these phenomena::

Table 7: Definition of Controversy Metrics

Dimension	Designation and Interpretation	Archetypal Examples
S1	**Fan-Carried Score:** Contestants with low judges' scores but high ranking due to popular vote.	Bobby Bones, Bristol Palin
S2	**Robbed Score:** Contestants with high technical merit but prematurely eliminated due to low votes.	Chandler Kinney, Mya

The computational formula for the **Fan-Carried Score** (S_1) is defined as follows:

$$S_1 = max(0, \overline{R}_{pct} - P_{pct}) \times (1 - \sigma_{norm}) \tag{29}$$

The specific methodologies for calculating the constituent variables are detailed below:

Table 8: Variable Calculation Methods

Variable	Definition	Formula
$\overline{R}_{pct}$	Average Judge Rank Percentile	$\frac{\overline{R}_{judge}-1}{N-1}$
P_{pct}	Final Placement Percentile	$\frac{P_{final}-1}{N-1}$
σ_{norm}	Normalized Std. Dev. of Weekly Judge Ranks	$\frac{\sigma_{rank}}{N-1}$
N	Number of Contestants in Season	-

The term $\overline{R}_{pct} - P_{pct}$ quantifies the magnitude of the rank disparity when a contestant's final standing significantly exceeds their technical ranking. The coefficient $(1 - \sigma_{norm})$ serves as a weighting factor for performance stability. Specifically, when a contestant demonstrates **consistently poor performance** (implying low variance in low scores), this coefficient increases, thereby amplifying the resultant controversy score.

The computational formula for the **Robbed Score** (S_2) is defined as:

$$S_2 = max(0, Z_{last}) \times max(0.5, 1 + \overline{Z}_{season}) \tag{30}$$

The methodologies for calculating the constituent variables of S_2 are detailed below:

Table 9: Definitions of Variables for S_2

Variable	Definition	Calculation Formula
Z_{last}	Terminal Week Z-Score	$\frac{Score_{last} - \mu_{week}}{\sigma_{week}}$
$\overline{Z}_{season}$	Seasonal Average Z-Score	$\frac{1}{W}\sum_{t=1}^{W} Z_t$

The term $max(0, Z_{last})$ quantifies the contestant's standardized performance during the specific week of their elimination. Controversy arises principally when a contestant is eliminated despite performing significantly above the weekly average. The factor $max(0.5, 1 + \overline{Z}_{season})$ evaluates the contestant's historical performance trajectory. If a consistently high-scoring contestant is eliminated, the magnitude of the controversy is amplified. Notably, should a contestant win the championship, S_2 is forcibly constrained to 0, as the concept of "unjust elimination" is definitionally inapplicable in such cases.

4.3.2 Case Studies

We applied the metrics mentioned above to calculate the controversy scores for all contestants.

It is evident that, with the exception of Billy Ray Cyrus, the S_1 scores of the other three controversial contestants mentioned in the problem are all very high, and their controversy rankings are also very prominent. Meanwhile, Billy Ray Cyrus ranks approximately in the top 10%. All of this demonstrates the rationality of the S_1 metric. The figure also marks the person with the highest S_2 score.

To determine whether using different rules would affect the final results, we conducted a simulation. Based on the popularity parameters inverted via SMC, we performed 1,000 Monte Carlo simulations to obtain the average final rankings.

For high S_1 contestants initially using the Percentage Rule:

Table 10: Performance of High S_1 Contestant under Different Rules

Contestant	S1	Outcome(Pct)	New Outcome (Rank)	Difference
Bobby Bones	0.500	1.9	2.8	+0.9
Bristol Palin	0.263	4.7	5.8	+1.1
Billy Ray Cyrus	0.141	5.5	6.2	+0.7

For these contestants, their rankings all declined after adopting the Rank Rule. These contestants' performance relies heavily on fan voting; when the influence of fans is limited by the Rank Rule, their results worsen.

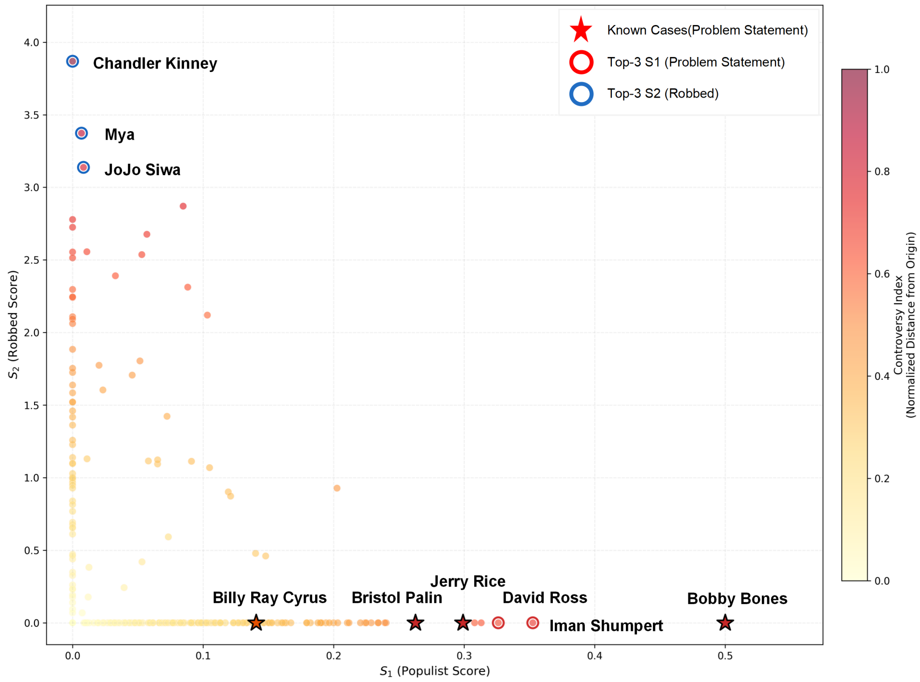

Figure 6: Controversy Quadrant

For high S_1 contestants initially using the Rank Rule:

Table 11: High S_1 Contestants Initially Using Rank Rule

Contestant	S1	Outcome(Rank)	New Outcome (Pct)	Difference
Jerry Rice	0.299	3.1	2.6	-0.5

Jerry Rice already possesses high popularity; switching to the Percentage Rule actually strengthens the influence of fan votes, resulting in a higher final ranking.

Further analysis of the top 10% of contestants ranked by S_1 reveals that the average ranking under the Rank Rule is 0.16 positions higher than under the Percentage Rule, and it yields a better ranking in 33.0% of cases (compared to only 22.5% for the Percentage Rule). The use of the Percentage Rule does not significantly improve the rankings of these contestants; this is likely because the contestants in the top 10% of S_1 are not exclusively "superstars," and their fan vote volumes are not always sufficient to significantly influence the final outcome.

We also utilized S_2 to identify two controversial contestants who received high Judge Score but were ultimately eliminated. Chandler Kinney delivered a dominant performance throughout the season yet finished in third place. Mya, a recognized technical perfectionist, lost to Donny Osmond.

Table 12: Analysis of Contestant with High S_2 Score

Contestant	Season	S_2 Score	Metric Rank	Original Rule	Change of Rank
Chandler Kinney	S33	3.869	#1	Rank	+0.5
Mya	S9	3.373	#2	Percentage	+0

It is evident that after applying the new rule, Chandler Kinney's result becomes worse. This is because her advantage in technical scores is diluted, while her disadvantage in fan votes is amplified. In contrast, Mya's result shows little change, likely because Mya possesses a more balanced foundation of both technical skills and fan support.

Further analysis of the top 10% of contestants ranked by S_2 reveals that their average ranking declines by 0.32 when the Percentage Rule is adopted. This indicates that this category of controversial contestants is more susceptible to the influence of rule changes.Simultaneously, to analyze whether the Judge Save impacts the results, the following analysis was conducted.

Table 13: Analysis of the Effect of Judge Save

Contestant	**S1 or S2**	**Rank (No Save)**	**Rank (with Judge Save)**	**Change of Rank**
Jerry Rice	S1: 0.299	3.1	3.6	+0.5
Bobby Bones	S1: 0.500	2.8	2.3	-0.5
Chandler Kinney	S2: 3.869	6.2	6.2	0.0

For these three cases, the effect of the **Judge Save** is not significant. Analysis of the top 10% of contestants ranked by and indicates that the Judge Save is not important for contestants with high scores; due to their large fan base, their probability of entering the **Bottom 2** is only 5.8%. In contrast, it is very important for contestants with high scores, who have a 28.4% probability of entering the **Bottom 2**.

4.4 Recommendation of Rule

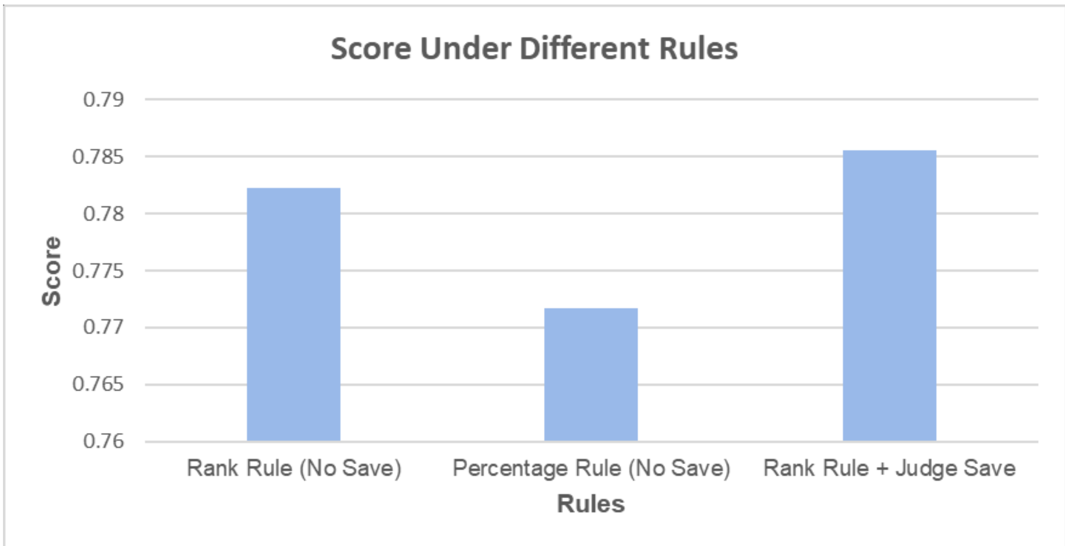

Figure 7: Comparison of Different Rules' Score

A good rule should comprehensively consider the opinions of both fans and judges. The following formula is used to calculate the optimal scheme:

$$\text{Score}(R) = w_{fan} \cdot \underbrace{\min(\text{Align}_{fan}(R), 0.75)}_{\text{Capped Fan Utility}} + w_{judge} \cdot \text{Align}_{judge}(R) \tag{31}$$

Here, w_{fan} is the weight for $Align_{fan}$, and w_{judge} is the weight for $Align_{judge}$. The setting $w_{judge} = 0.7 > w_{fan} = 0.3$ indicates that, as a dance competition show, Meritocracy should occupy the dominant position in determine the outcomes. Simultaneously, 0.75 is set as a threshold for fans' influence; extremely high values exceeding this threshold typically imply Populism Overflow and should not receive additional system rewards. A similar practice is found in the Companies Act 2006, s.21[5] , and George Tsebelis (2002) also theoretically demonstrates the Rationale behind using this value[6].

Based on the metric formula above, the scores for different rules were calculated.

Table 14: Final Score under Different Rules

Rules	Score
Rank Rule (No Save)	0.782
Percentage Rule (No Save)	0.772
Rank Rule with Judge Save	**0.785**

Rank Rule with Judge Save is the optimal rule, as it achieves a better balance between technical merit and popularity. Simultaneously, the Judge Save mechanism enables judges to "eliminate" or "save" borderline contestants, which prevents the occurence of certain extreme situations.

5 Task 3: Quantification of Influential Factors

5.1 Problem Analysis

Objective 1: Analyze exactly how significant the impact of Professional Partners is on competition outcomes. Do they merely provide technical guidance, or do they directly boost fan interest?

Objective 2: Determine whether the features prioritized by judges and fans are identical, and whether the importance of various contestant attributes (e.g., age, gender, occupation) are consistent across the two evaluation systems.

This is a problem involving **Causal Inference on Observational Data** and **Multilevel Variance Decomposition**. It requires establishing models to explain both audience ratings and judges' scores.

5.2 Model Setting

To precisely isolate the contributions of each explanatory variable, we constructed a **Triple-Track Mixed-Effects Model (3T-MEM)**. This model utilizes three independent and parallel Linear Mixed-Effects Model (LMM) tracks to model distinct feature dimensions or variable groups within the complex dataset, subsequently fusing information across the tracks. This results in more precise parameter estimation and data analysis.

5.2.1 Defination of Variables

In the **Triple-Track Mixed-Effects Model (3T-MEM)**, Track A corresponds to the judge evaluation component. The dependent variable for Track A is defined as . Track B corresponds to the fan evaluation component. The dependent variable for this track is $Y_{i,t}^{F} = \text{logit}(\hat{\pi}_{i,t})$ ..

The contestant feature vector X_{Traits} includes the standardized contestant age defined as X_{Age}, the categorical variable for the contestant's industry defined as $X_{Industry}$, and the competition week defined as X_{Week}. To capture the long-term, specific influence of "partner identity," we select the Professional Partner random intercept u_{Pro}. To address the autocorrelation among multiple observations of the same contestant, we select the Star individual random intercept u_{Star}.

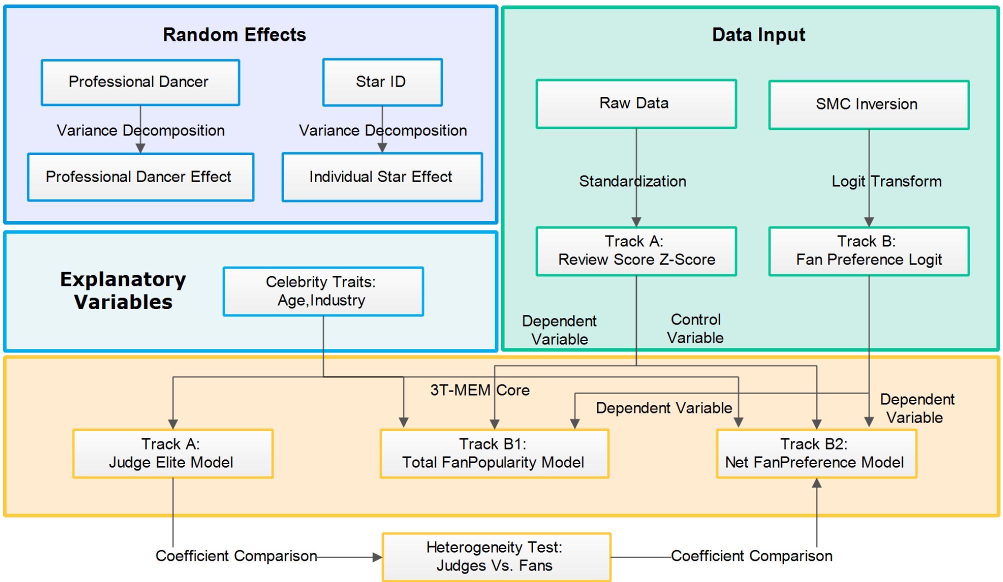

Figure 8: 3T-MEM Flowchart

5.2.2 Modelling

For Track A, to dissect the judges' scoring logic and verify whether it is purely based on technical performance, we construct The Meritocracy Model. The specific calculation formula is as follows

$$Y_{i,t}^{J} = \beta^{J} \cdot X_{Traits} + u_{Pro}^{J} + u_{Star}^{J} + \epsilon \tag{32}$$

For Track B1, to dissect the overall voting logic of the fans and reflect their general favorability toward contestants—incorporating both dancing skills and individual charisma—we construct The Fan Popularity Model. The specific calculation formula is as follows:

$$Y_{i,t}^{F} = \beta^{F1} \cdot X_{Traits} + u_{Pro}^{F1} + u_{Star}^{F1} + \epsilon \tag{33}$$

For Track B2, to dissect the pure fan preference after stripping away technical performance, we innovatively introduce the current week's judge scores Y^{J} into the regression equation as a control variable, thereby isolating the pure popularity bias. We construct The Fan Bias Model. The specific calculation formula is as follows:

$$Y_{i,t}^{F} = \gamma \cdot Y_{i,t}^{J} + \beta^{F2} \cdot X_{Traits} + u_{Pro}^{F2} + u_{Star}^{F2} + \epsilon \tag{34}$$

Here, the coefficient β^{F2} represents pure "likability" or "sympathy votes," no longer encompassing the portion of votes earned due to superior performance.

5.2.3 Model Diagnostics: VIF Check

To ensure the independence of each explanatory variable, we calculated the **Variance Inflation Factor (VIF)**. The calculation formula is as follows:

$$\text{VIF}_j = \frac{1}{1 - R_j^2} \tag{35}$$

Where R_j^2 is the coefficient of determination obtained from an auxiliary regression treating the j-th variable as the dependent variable and the remaining variables as independent variables.

Table 15: VIF Check Results

Variable	VIF	Conclusion
Judge Score Z	1.27	No Multicollinearity
Age (Standardized)	1.44	No Multicollinearity
Industry (Avg)	~ 1.10	No Multicollinearity

To answer the question "how significant is the impact of Professional Partners on the results," we employed numerical algorithms to solve the model parameters, thereby calculating the Intraclass Correlation Coefficient (ICC). The ICC value reflects the proportion of the total variance explained by the "Professional Partner" variable. Its mathematical definition is as follows:

$$\mathrm{ICC}_{Pro} = \frac{\sigma^2_{Pro}}{\sigma^2_{Pro} + \sigma^2_{Star} + \sigma^2_{\epsilon}} \tag{36}$$

Where σ^2_{Pro} represents the variance of the Professional Partner random effects, and the denominator represents the total residual variance (including the Star individual variation σ^2_{Star} and σ^2_{ϵ}).

Table 16: Decomposition of Professional Partner Variance Contribution (Pro ICC)

Track	Dependent Variable	Pro ICC (ρ_{Pro})
Track A	Judge Score	< 0.001 (Negligible)
Track B1	Fan Vote Share	< 0.001 (Negligible)
Track B2	Fanbias	0.0029 (Low)

Conclusion:the direct statistical explanatory power of Professional Partners on competition outcomes is close to zero.

5.3 Result Analysis: Heterogeneity of Evaluation Systems

To determine whether the features prioritized by judges and fans are identical, we compared the regression coefficients from Track A (Judge) and Track B2 (Fan Bias), revealing significant structural differences.

Industry: To deeply dissect the differential treatment of contestants from distinct backgrounds, we extracted the regression coefficients for all industries. We calculated the coefficient difference $\Delta = \beta^{F2} - \beta^{J}$.When$\Delta > 0$, fans are more lenient than judges (or judges are too strict).We selected nine industries for analysis.

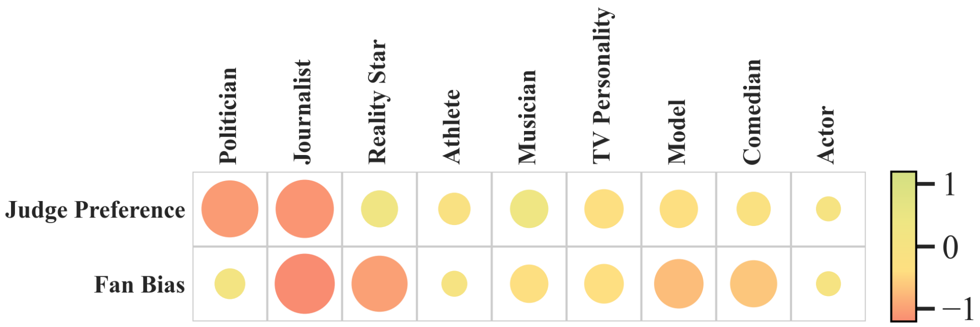

Figure 9: Industry Effects Illustration

Table 17: Industry Effects: Judge vs. Fan Bias

Industry	Judge's Coef β^J	Fan Bias β^{F2}	Difference Δ
Politician	-1.05*	+0.13 (ns)	+1.18
Journalist	-1.11	-1.20	-0.09
Reality Star	+0.30	-1.01*	-1.31
Athlete	-0.17**	+0.02	+0.19
Musician	+0.34	-0.34	-0.68
TV Personality	-0.37***	-0.38***	-0.01
Model	-0.34**	-0.73***	-0.39
Comedian	-0.22	-0.64***	-0.42
Actor	0.00	0.00	0.00

*explanation:**** $p < 0.001$,** $p < 0.01$,* $p < 0.05$,. $p < 0.1$,ns, meaning that it's not significant。

Due to the large sample size, the p-value can be calculated using the following formula.

$$Z = \frac{\hat{\beta}}{SE(\hat{\beta})}, \quad p = 2(1 - \Phi(|Z|)) \tag{37}$$

It is evident that relative to **Actors**, the average scores for **TV Personalities** and **Models** are significantly less than 0, indicating their average performance is inferior to that of Actors. This may be attributed to the relatively superior technical skills of Actors. **Athletes** are significantly inferior to Actors only in the judges' evaluation; this is likely because, compared to fans, judges place greater emphasis on technical proficiency, and athletes are significantly inferior to professionally trained actors in terms of physical expressiveness. Meanwhile, fans show a clear preference for **Politicians**, which may be related to their high public influence despite their average technical skills. Conversely, judges demonstrate a stronger preference for **Reality Stars** and **Musicians**.

Age: The coefficient of the Judge Model (β^J_{Age}) is -0.42, compared to -0.24 in the Fan Bias Model (β^{F2}_{Age}). It is evident that although both sides favor younger contestants, fans demonstrate a significantly higher tolerance for older contestants than the judges.

6 Task 4: System Design

6.1 ACDW-B3 Modelling

We propose the ACDW-B3 system to balence fan engagement with professional meritocracu. ACDW-B3 stands for **Adaptive Concave Diminishing-returns Weighted Bottom-3 System**. Using the metrics constructed earlier in 4.4, this system achieves a final score of 0.9053, outperforming the previously identified optimal solution, the Rank Rule with Judge Save (Score = 0.7827).

Step A: Fan Influence Dilution

To address the issue of excessive fan voting influence under the Percentage Rule, we map the raw fan vote share $\pi_{i,t}$ into a non-linear utility space, thereby making fan vote to have a diminishing marginal utility.

$$f_{i,t} = \frac{(\pi_{i,t})^p}{\sum_{k \in C_t} (\pi_{k,t})^p}, \quad p \in (0,1) \tag{38}$$

Where parameter p (compression factor) controls the degree of compression. We set $p = 0.55$.

Step B: Consensus-Driven Weighting

we constructed a Consensus Meter to determine the weights on jugde's score and fan vote share. We define ρ_t as the Spearman correlation coefficient between the judges' rankings and the fan vote rankings for the current week. The judges' weight λ_t is dynamically adjusted according to ρ_t :

$$\lambda_t = \lambda_{min} + (\lambda_{max} - \lambda_{min}) \cdot \frac{1 - \rho_t}{2} \tag{39}$$

High Consensus ($\rho \approx 1$), $\lambda_{min} = 0.6$. When everyone agrees on who danced well, allow fans to determine the minor ranking differences (Pro-Fan).

Low Consensus ($\rho < 0$), $\lambda_{max} = 0.8$: When the judges and the audience hold diverging opinions, the system increases the judges' weight to safeguard professional integrity.

Step C: Comprehensive Score and Bottom-3 Mechanism

Bottom-3 Danger Zone: The three contestants with the lowest score $S_{i,t}$ enter the danger zone. Compared to the traditional Bottom-2, expanding the danger zone increases the show's suspense (Drama) and affords the judges greater discretionary latitude.

$$S_{i,t} = \lambda_t \cdot \frac{J_{i,t}}{\sum J} + (1 - \lambda_t) \cdot f_{i,t} \tag{40}$$

Judge Save Protocol: The judges hold the final decision-making power within the Bottom-3 and must eliminate the contestant with the lowest technical score (J).

6.2 Model Evaluation

We calculate the metrics for the new rule R and compare it with the best rule identified earlier, Rank Rule with Judge Save.

Table 18: Multi-Dimensional Performance Comparison of Two Rules

Rule	**Fan Align**	**Judge Align**	**Score**
ACDW-B3	0.7630	0.9720	**0.9053**
Rank + Save	0.6570	0.8370	0.7827

The comprehensive score of ACDW-B3 has seen a substantial increase. This is primarily attributed to its astounding Judge Alignment (0.972)—it almost perfectly executes the professional will of the judges, avoiding Fan-Carried controverises. Meanwhile, the introduction of dynamic weighting allows the rule to maintain fairness in results during exciting moments of high disagreement between judges and fans.

7 Sensitivity Analysis

In Section 4.4, we set w_{judge} to 0.7. If we increase or decrease it by 0.05, the results are as follows:

Table 19: Sensitivity Analysis of w_{judge}

w_{judge}	Rank Rule + Judge Save	Rank Rule	Percentage Rule
0.7	0.7885(best)	0.7822	0.7717
0.65	0.7808(best)	0.7754	0.7701
0.75	0.7963(best)	0.7890	0.7733

It is evident that the parameter changes do not affect the final proposed scheme, indicating that the parameter settings are reasonable.

8 Strengths and Weaknesses

8.1 Strengths

- **Rigorous Counterfactual Framework:** Our model enables ”re-running” history to identify ”robbed” contestants and quantify directional biases ($Pro - Fan$ vs. $Capping$), providing a scientific basis for rule evaluation.
- **High Explanatory Power via 3T-MEM:** Our model effectively disentangles technical merit from pure popularity, resolving long-standing debates on ”Professional Partner” influence and industry-specific preference divergences.

8.2 Weaknesses

- **Simplification of Social Dynamics:** The assumed **Markovian Property** focusing on weekly momentum may underestimate multi-year fan loyalties or long-term celebrity archetypes that span across multiple seasons.

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Report on Use of AI

1. OpenAI ChatGPT (GPT-4o)

Query: My LaTeX document is producing "Overfull \hbox" warnings for long tables. I also need to configure the header to display "Team # 2607504" on the left. How can I address these issues?

Output: I suggested adding \setlength{\emergencystretch}{4em} to the preamble and using the fancyhdr package to configure the headers.

2. Google Gemini (Feb 1, 2026 version)

Query: Please refine this sentence for a formal research paper: "Fans tend to support celebrities for who they are, not solely for their dancing." Also, suggest a professional name for a scoring mechanism that uses a concave function.

Output:
- Refinement: "Competition outcomes are predominantly governed by the celebrities' intrinsic attributes, rather than solely by technical performance."
- Suggestion: "Adaptive Concave Diminishing-returns Weighted System (ACDW)."

2. Google Gemini (Feb 1, 2026 version)

Query 1: Please refine this sentence for a formal research paper: "Fans tend to support celebrities for who they are, not solely for their dancing." Also, suggest a professional-sounding name for a scoring mechanism that applies a concave function to reduce the influence of extreme fan votes.

Output:

- **Refinement:** "Competition outcomes are predominantly governed by the celebrities' intrinsic attributes and individual charisma, rather than solely by technical performance."
- **Suggested name:** Adaptive Concave Diminishing-returns Weighted System (ACDW).

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3. GitHub Copilot (Jan 20, 2026 version)

Query 1: (In-editor prompt) Write a Python function using pandas to calculate the Z-score for judges' scores within each season and handle missing values by filling them with the median of that specific season.

Output:

```
def normalize_scores(df):
    df['score'] = df.groupby('season')['score'].transform(lambda x: x.fillna(x.median()))
    df['z_score'] = df.groupby('season')['score'].transform(lambda x: (x - x.mean()) / x.std())
    return df

```